

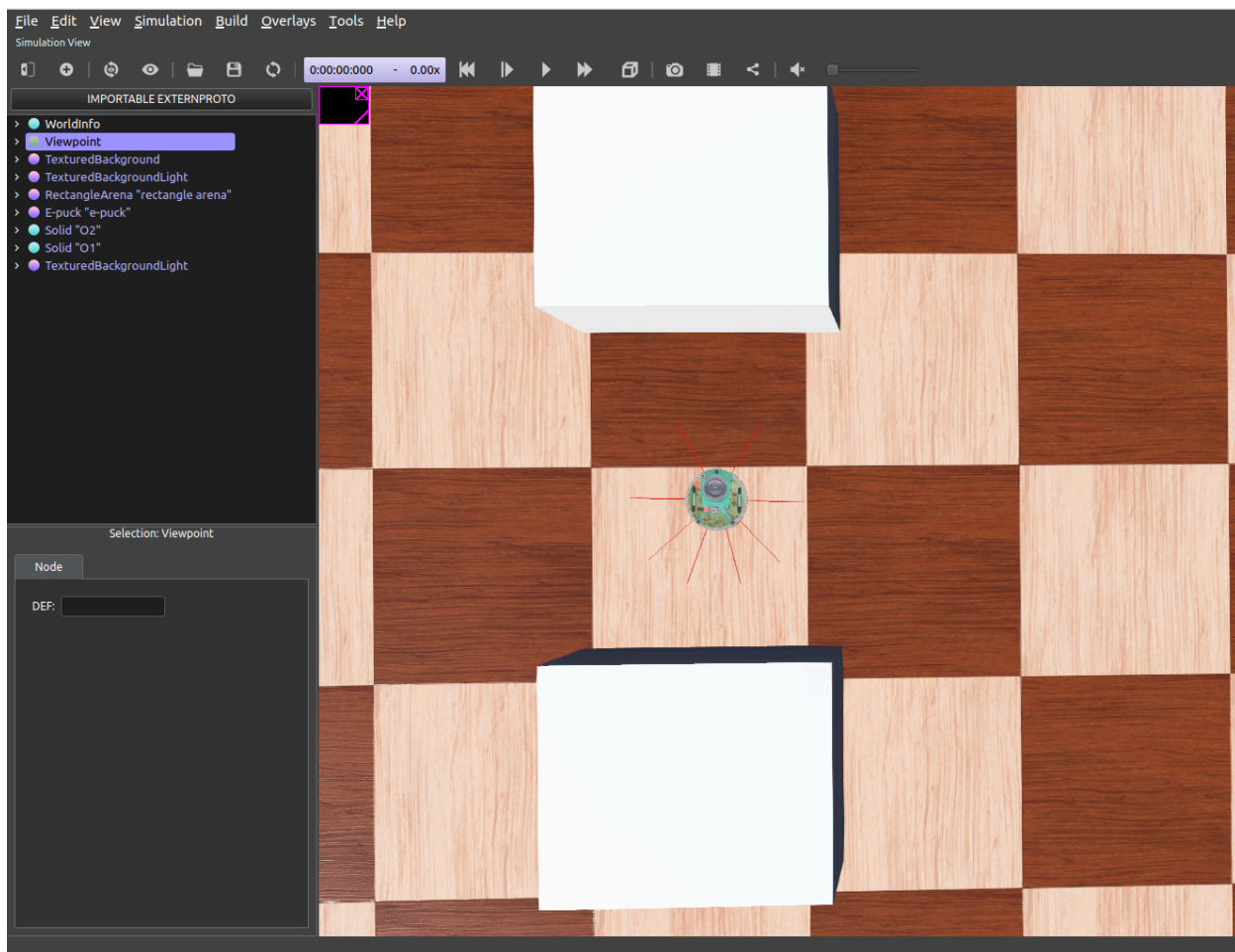
Assignment-3: Basic Reactive Behaviors

Introduction

In this assignment, a basic reactive control with pre-defined behaviour and a finite state machine is implemented.

World snapshot

The world contains two obstacles (**O1 box = Green**, **O2 box = Blue**) and an e-puck robot inbetween. As shown in the image below:



The controller is able to complete the actions required by the assignment with the finite state machine (FSM) below.

States in Finite state machine (FSM)

I implemented the assignment with a total of 5 states as shown below:

MOVING_FORWARD

A velocity of π (3.14) radian is set for both left and right motors equally to move forward. This state persists until an obstacle is detected in the front proximity sensors (**ps7** and **ps0**).

ROTATE_180

To rotate 180 degrees, a velocity of $-\pi$ is set to the left motor, and a velocity of $+\pi$ is set to the right motor, for a timespan of ~ 1.5 seconds. Once the rotation is completed, state is assigned as MOVING_FORWARD.

ROTATION_90_RIGHT

Once the second obstacle is found, the robot is rotated 90 degrees to the right. To rotate 180 degrees, a velocity of $-\pi$ is set to the left motor, and a velocity of $+\pi$ is set to the right motor, for a timespan of ~ 0.75 seconds.

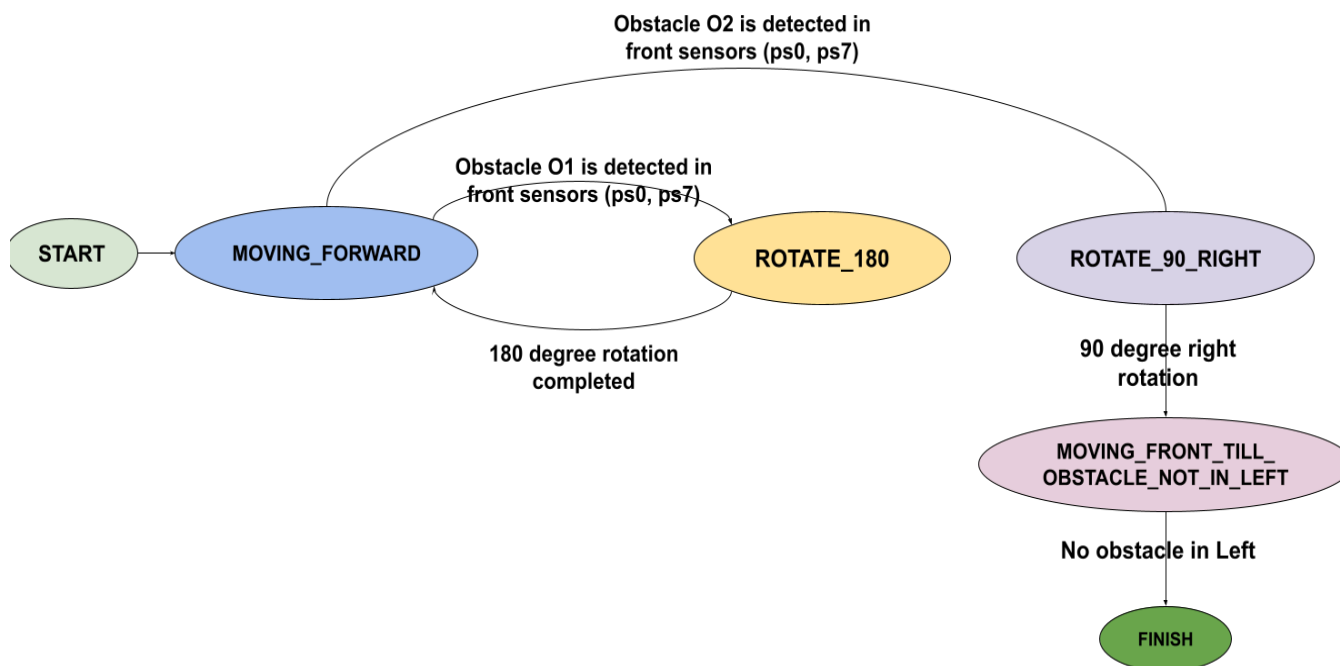
MOVE_FRONT_UNTIL_OBSTACLE_NOT_IN_LEFT

Once the 90 degree right rotation is completed, the robot is assigned with MOVE_FORWARD_UNTIL_OBSTACLE_NOT_IN_LEFT state, where the left proximity sensor (**ps5**) is observed until the proximity value is less than a threshold.

FINISH

When the obstacle is not in the left side, the robot is stopped by assigning 0 velocity to both wheel motors.

FSM diagram



Conclusion

With the above FSM, the robot can complete the following actions:

1. Go forward until Obstacle O1
2. When the obstacle is detected in front proximity sensors (**ps7** and **ps0**), it turns 180 degree
3. Moves forward until obstacle O2 is detected in its front sensors
4. Turns 90 degrees right
5. Move forward until the obstacle is not in the left of the robot
6. STOP after the robot passes the left obstacle and the left sensor (ps5) does not detect an obstacle.